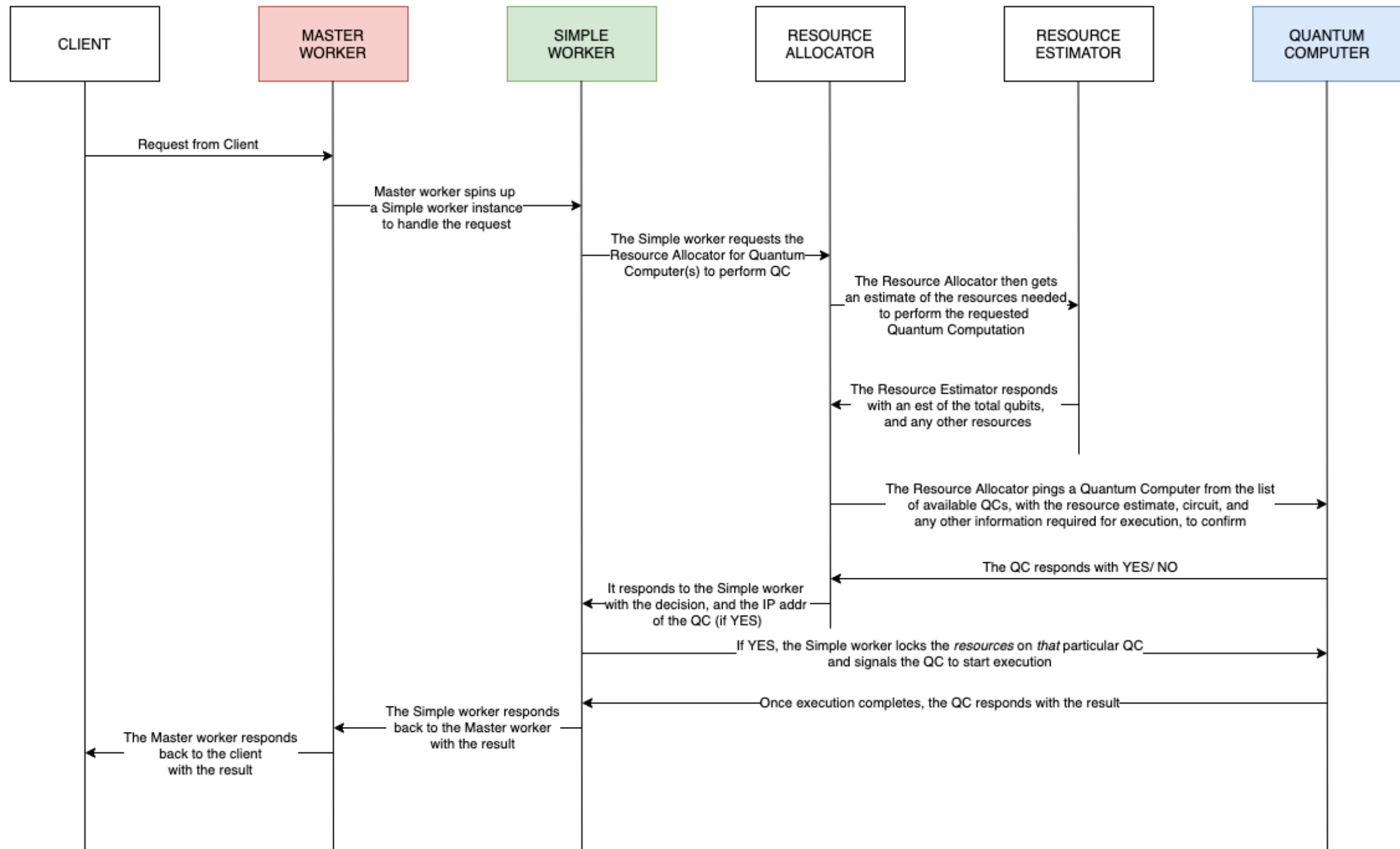


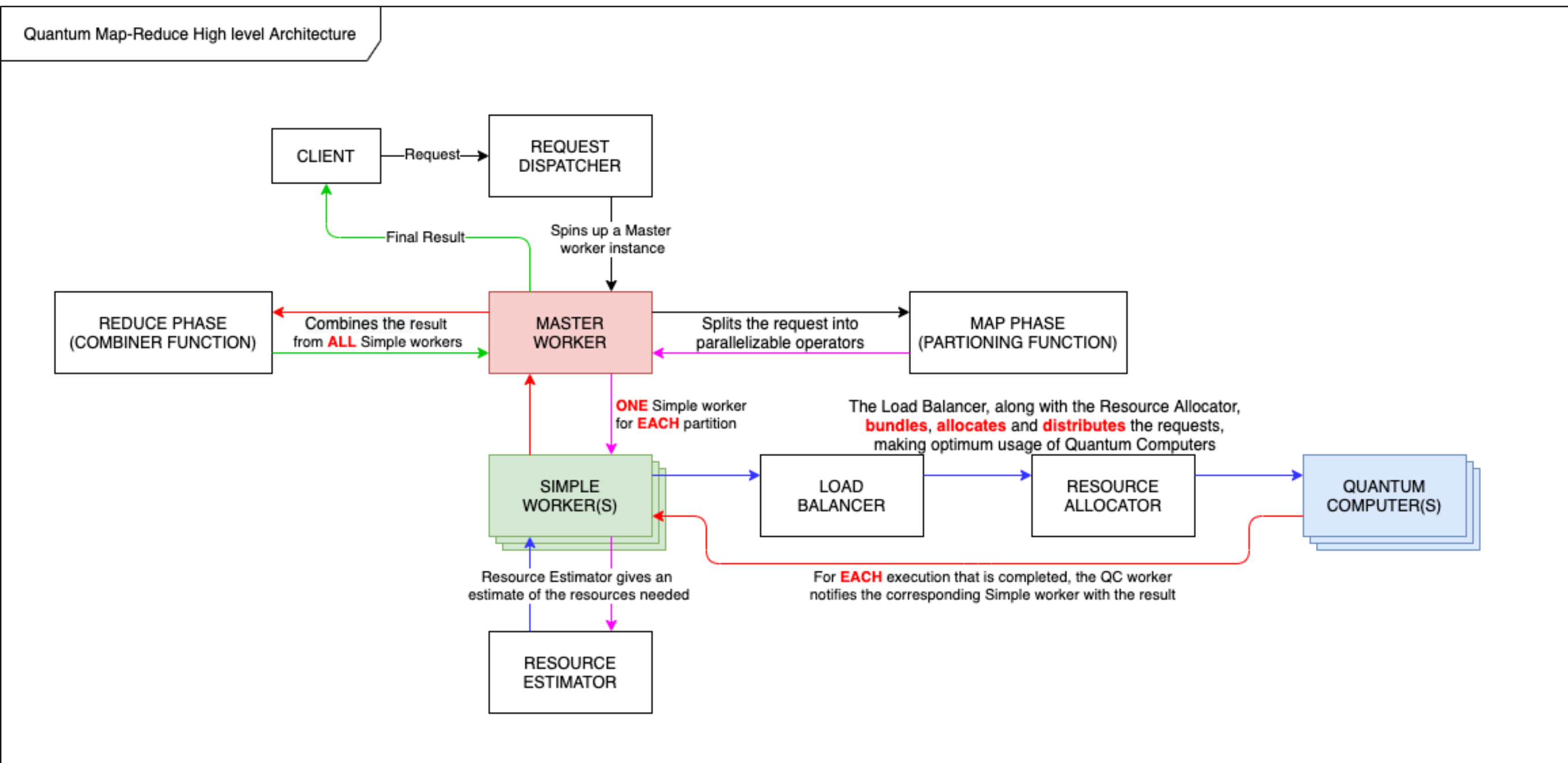
Distributed Quantum Computing

High level Sequence diagram

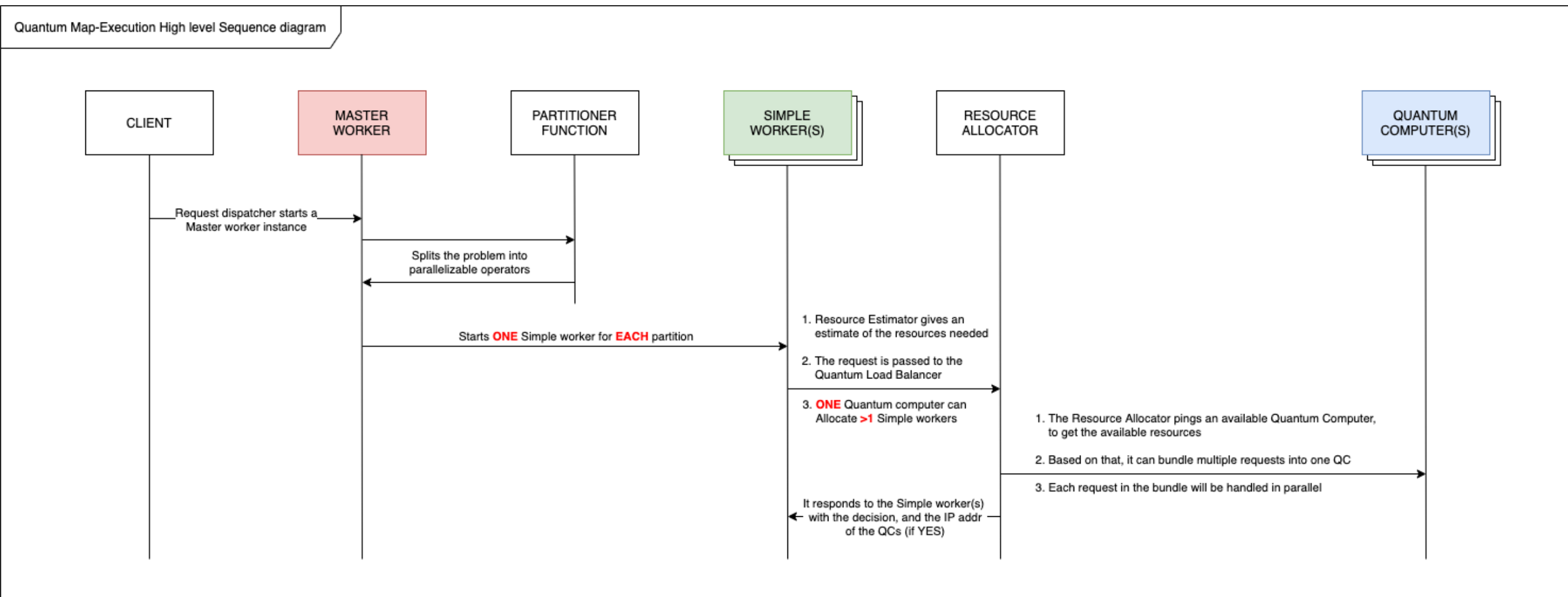
High level Sequence diagram



Quantum Map-Reduce High level Architecture

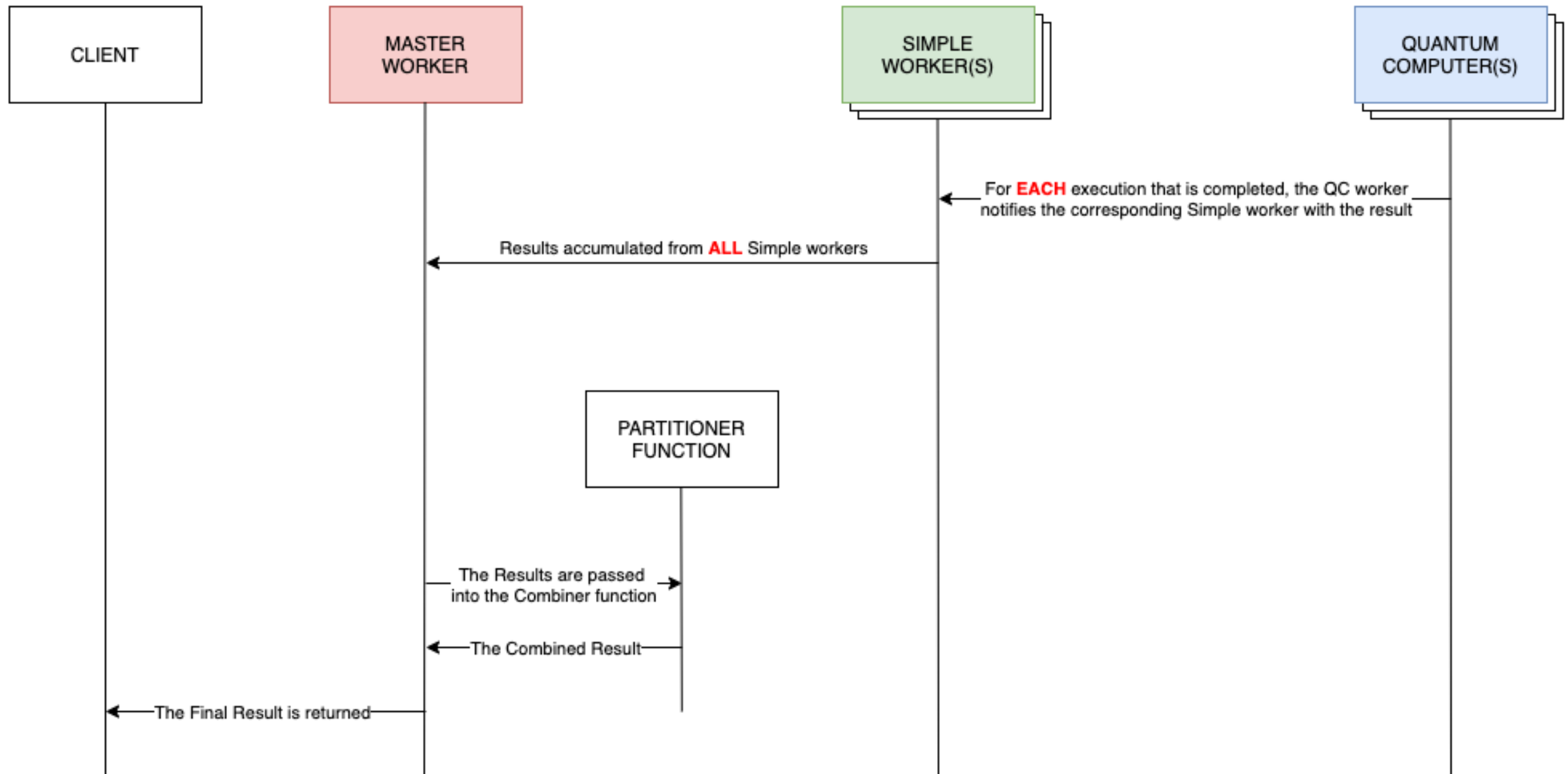


Quantum Map-Execution High level Sequence diagram



Quantum Execution-Reduce High level Sequence diagram

Quantum Execution-Reduce High level Sequence diagram



Map-Reduce functions

Map

- Partition function, based on the Quantum algorithm
- Ordering — Sequential or Parallel
- Optimization

Reduce

- Combine function, based on the Quantum Algorithm

Assumptions

1. No distribution of qubits between multiple Quantum Computers. We will assume that all quantum resources (qubits) required to execute a quantum circuit, will be accommodated on ONE computer
2. We will start with requests having only QASM instructions, and later expand on accepting circuits, and conversions of those into QASM
3. No optimization for now

Interconnected components

- Quantum Algorithms
- Partition type
- Partition ordering (Sequential or Parallel)
- Internal distributed traffic
- Load Balancer algorithm
- Task Scheduler algorithm
- Physical layout of Quantum Computers